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DCYBER TECHLAB PVT.LTD

Project Report:

Adversary Emulation and Detection Validation: A Purple Team Framework

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1. Executive Summary

Traditional cybersecurity often relies on passive monitoring, assuming that security tools (SIEM/EDR) are working correctly "out of the box." This project challenged that assumption by building an active "Purple Team" framework. By establishing a controlled lab environment and simulating specific MITRE ATT&CK tactics (Red Team), I was able to validate the detection capabilities of the Wazuh SIEM (Blue Team). The project successfully demonstrated that default configurations often miss critical threats and that continuous detection engineering is required to close visibility gaps.

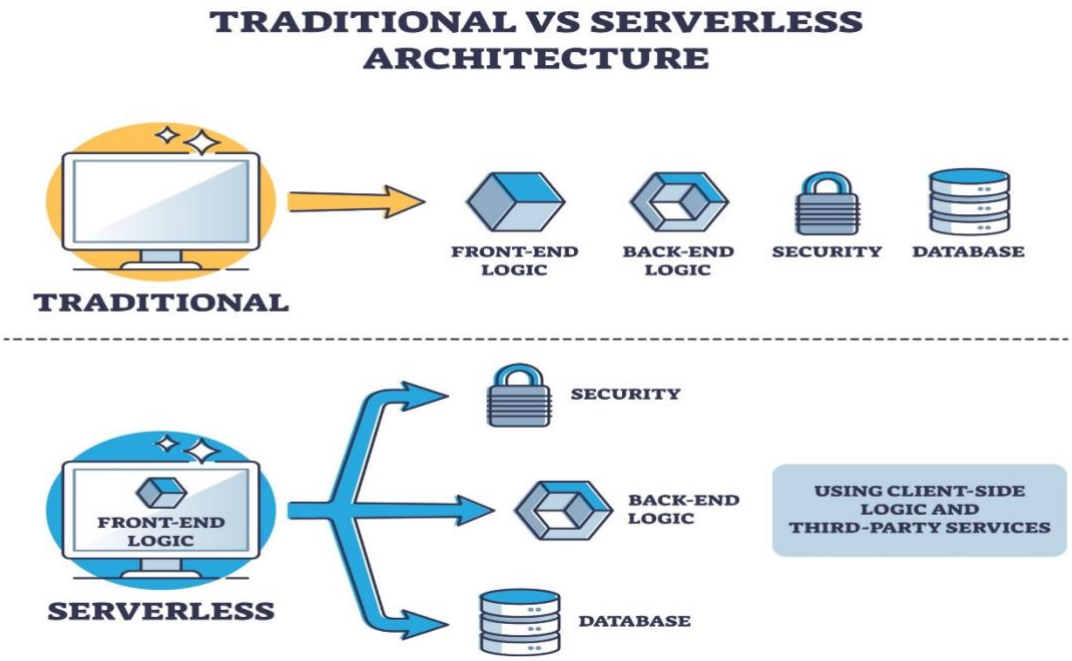
2. Problem Statement

Modern organizations invest heavily in security tools, but rarely validate them against real-world attack behaviors. Relying on default rule sets leaves "blind spots" where advanced adversaries can hide. There is a critical need for a verified feedback loop that simulates realistic attacks to proactively tune defence systems before a real breach occurs.

3. Infrastructure Setup

To conduct this testing safely, an isolated virtual lab was constructed:

- **Hypervisor:** Oracle VirtualBox.
- **Network Configuration:** Custom Host-Only Network (192.168.56.x) to ensure air-gapped communication between VMs, plus a NAT adapter for package updates.
- **Victim Node:** Windows 11 Enterprise.
    - *Configuration:* Enabled Process Creation Auditing (Event ID 4688) and File Integrity Monitoring.
- **Attacker/SIEM Node:** Kali Linux.
    - *Software:* Hosted the Wazuh Manager (v4.7.5) and Elastic Stack components.
- **Tooling:**
    - **Atomic Red Team:** For adversary emulation.
    - **Wazuh Agent:** For endpoint telemetry collection.
    - **Sysmon/Auditpol:** For granular Windows event logging.



4. Methodology: The Purple Team Loop

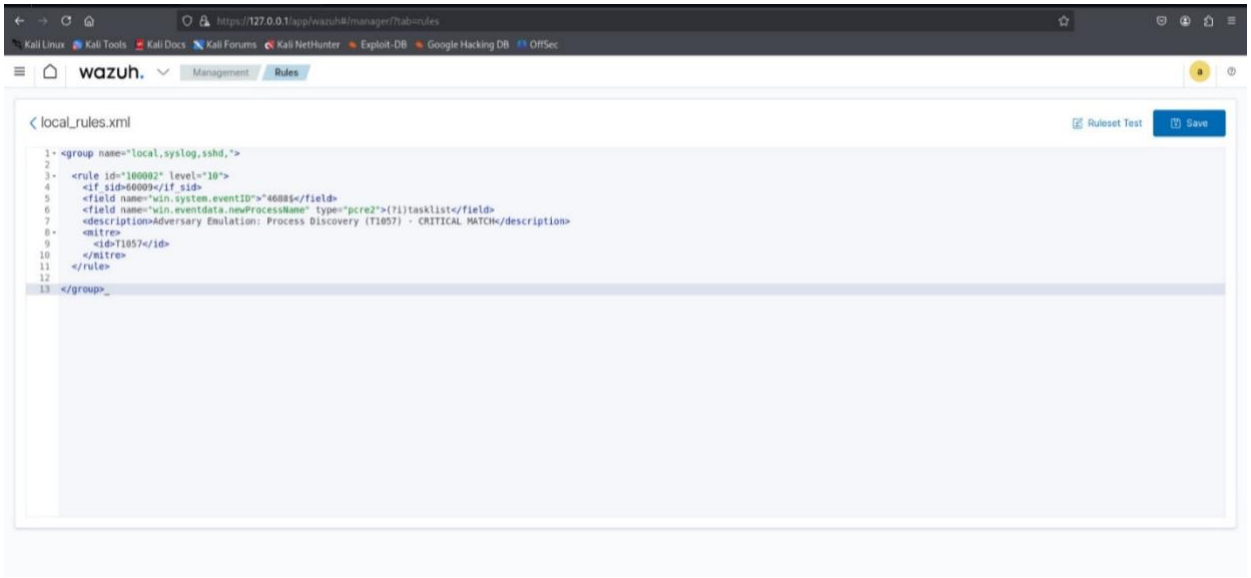
The project followed a cyclical methodology for three distinct attack types:

1. **Emulate (Red):** Execute a specific technique using Atomic Red Team.
2. **Hunt (Blue):** Analyze raw logs (archives.json and Event Viewer) to confirm visibility.
3. **Detect (Purple):** Engineer and tune custom XML detection rules to generate high-fidelity alerts.

5. Execution & Analysis

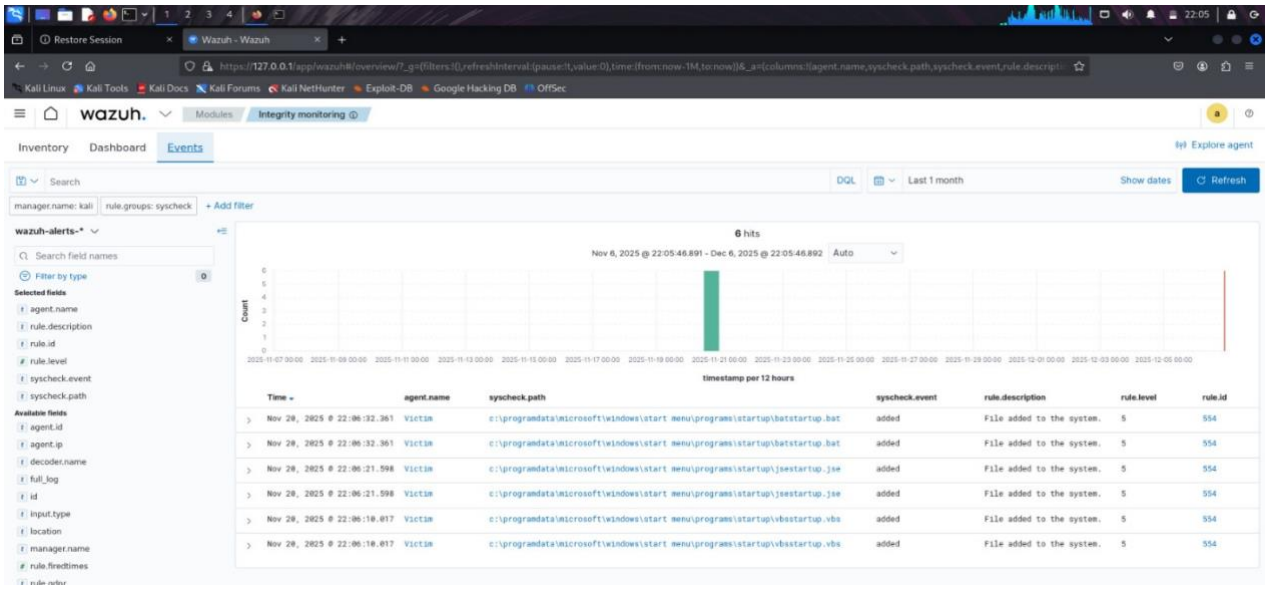
Cycle 1: Discovery (MITRE T1057)

- **Attack Scenario:** An adversary attempting to list running processes to identify antivirus or security tools.
- **Execution:** Ran Invoke-AtomicTest T1057 which executed tasklist.exe.
- **Detection Engineering:**
    - Initially, the log was received by the Wazuh Manager but did not generate an alert due to low default severity scores.
    - **Solution:** Created a custom XML rule (ID 100002) targeting Event ID 4688 with a regex match for tasklist.exe
    - **Outcome:** Successfully generated a **Level 10 Severity Alert**.



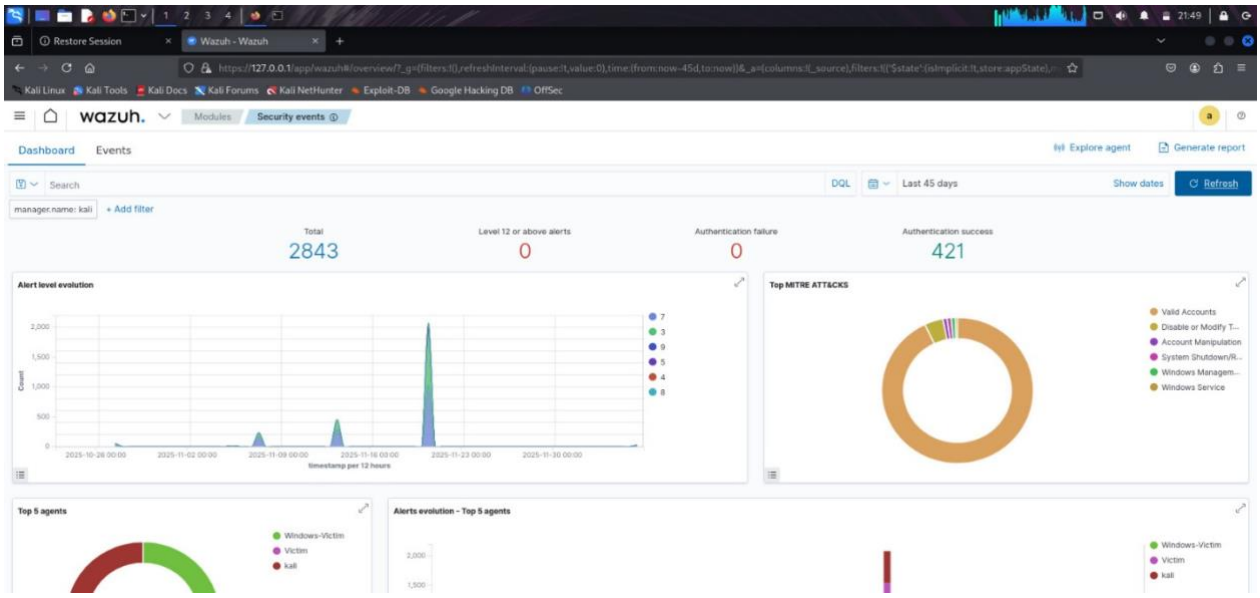
Cycle 2: Persistence (MITRE T1547)

- **Attack Scenario:** An adversary attempting to maintain access across reboots by modifying the Registry and Startup folder.
- **Execution:**
  - Dropped malicious scripts (batstartup.bat) into the Windows Startup folder.
  - Created a Registry Run Key pointing to a fake malware payload (EvilMalware).
- **Detection:**
  - Utilized Wazuh's **File Integrity Monitoring (FIM)** module.
  - **Outcome:** The system successfully detected the file creation, triggering **Rule 554 (File added to the system)**. This validated that the SIEM can detect unauthorized persistence mechanisms without needing custom rules.



Cycle 3: Credential Access (MITRE T1003)

- **Attack Scenario:** Attempting to dump the memory of the LSASS.exe process to steal user passwords and Kerberos tickets.
- **Execution:** Used rundll32.exe and comsvcs.dll to attempt a memory dump.
- **Defence Validation:**
    - Windows Defender Tamper Protection actively blocked the attack.
    - **Outcome:** The system generated "Access Denied" and "Malicious Script Detected" logs. This validated that the endpoint protection layer was active and effective, even if the SIEM did not receive a "Process Creation" log for the blocked attempt.



6. Key Technical Challenges & Solutions

This project involved significant troubleshooting of real-world engineering problems:

1. **Timestamp Synchronization (The "Invisible Log" Issue):**
    - ***Problem:*** Logs arriving from the Windows VM were being discarded or hidden by the Dashboard because the VM's clock was 10 hours behind the Wazuh Manager.
    - ***Solution:*** Manually synchronized the VM clocks and adjusted Dashboard time filters to "Last 24 Hours" to locate historical data.
2. **Rule Shadowing:**
    - ***Problem:*** Custom rules were initially ignored because default system rules took precedence.

- ***Solution:*** Utilized the overwrite="yes" tag in the XML configuration and adjusted rule levels to ensure the custom logic was prioritized by the analysis engine.

3. **Pipeline Debugging:**
    - ***Problem:*** Differentiating between "Log not sent" and "Alert not generated."
    - ***Solution:*** Mastered the use of wazuh-logtest for logic verification and grep commands on archives.json vs alerts.json to trace the data flow through the SIEM pipeline.

7. Future Scope

While this project established a robust detection framework for Windows-based threats, the Purple Team methodology can be expanded in several key areas:

- **Automated Response:** Implementing Wazuh **Active Response** scripts to automatically block IP addresses or kill malicious processes (like rundll32.exe) immediately upon detection, moving from "Detection" to "Prevention."
- **Linux & Cloud Coverage:** Expanding the atomic testing scope to include Linux servers and Cloud workloads (AWS/Azure), specifically testing for SSH persistence and S3 bucket enumeration.
- **Dashboards & Reporting:** Developing custom Kibana visualizations to track "Mean Time to Detect" (MTTD) metrics, providing leadership with a quantifiable view of security posture improvements over time.

8. Conclusion

This project successfully transformed a passive lab into an active detection engineering environment. By moving through the cycle of **Attack -> Analyze -> Detect**, I demonstrated that security is not a product but a process. The final framework allows for continuous validation of new threats, ensuring that defences are not just theoretical, but proven.

9. Acknowledgement

I would like to express my sincere gratitude to my mentor, Mr. Pruthvi Chauhan, Associate Consultant at DCyber Techlab, for his invaluable guidance and support throughout the course of this project. His deep insights into cybersecurity operations and his practical advice on detection engineering were instrumental in the successful completion of this Purple Team framework.

I am also grateful for the opportunity provided by DCyber Techlab to work on such a challenging and relevant topic, which has significantly enhanced my technical skills and understanding of the cyber defence landscape.